

Required and optional software for the shortcourse



- <https://julialang.org/downloads/>
 - Allow to install and use MAGEMinApp and MAGEMin_C
- Visual Studio Code – Code editor <https://code.visualstudio.com/>
 - Allow Windows user to connect to Linux WSL and get the best performance for MAGEMin
- <https://inkscape.org/fr/>
 - Allow to open MAGEMinApp figure vector format (svg)



- **WINDOWS users**
- Recommended to install and use WSL (Windows sub-system for Linux)
- Connect to WSL and Linux Via VSCode (see next slides)
- MAGEMin performs twice faster on Linux/Mac than native Windows

Julia installation

Mac and Linux

Julia installation MAC LINUX

Install

Install the latest Julia version ([v1.10.5](#) August 27, 2024) by running this in your terminal:

```
$ curl -fsSL https://install.julialang.org | sh
```

COPY

For Windows instructions [click here](#)

Once installed `julia` will be available via the command line interface.

This will install the `Juliaup` installation manager, which will automatically install julia and help keep it up to date. The command `juliaup` is also installed. To install different julia versions see `juliaup --help`.

Please star us [on GitHub](#). If you use Julia in your research, please [cite us](#). If possible, do consider [sponsoring](#) us.

```
curl -fsSL https://install.julialang.org | sh
```

Julia installation

Windows

Julia installation WINDOWS

Install

Install the latest Julia version ([v1.10.5](#) August 27, 2024) from the [Microsoft Store](#) by running this in the command prompt:

```
> winget install julia -s msstore
```

COPY

It looks like you're using Windows. For Linux and MacOS instructions [click here](#)

Once installed `julia` will be available via the command line interface.

This will install the [Juliaup](#) installation manager, which will automatically install julia and help keep it up to date. The command `juliaup` is also installed. To install different julia versions see `juliaup --help`.

Please star us [on GitHub](#). If you use Julia in your research, please [cite us](#). If possible, do consider [sponsoring](#) us.

```
winget install julia -s msstore
```

MAGEMinApp & MAGEMin_C installation

Windows
Mac and Linux

Install MAGEMinApp

- Open a new terminal (vscode for those who use it, or terminal/powershell for others)
- Type:
julia -t 6 # where 6 is the number of core you want to use (depends on your machine, type
 # 'versioninfo()' to get more informations)

```
seph42@DESKTOP-2V82075:~$  
seph42@DESKTOP-2V82075:~$ julia -t 6  
  
      _  
   _(_)_  
  / __ \ |  
 | | ___| | |
 | || __ \| |  
 | |_|___) | |  
 |  __/ | |  
 |_/_____|_|  
          |  
Documentation: https://docs.julialang.org  
Type "?" for help, "]?" for Pkg help.  
  
Version 1.10.0 (2023-12-25)  
Official https://julialang.org/ release  
  
(v1.10) pkg> add MAGEMinApp
```

- In the terminal type ']', this will open the package manager
- Type 'add MAGEMinApp', this will download and install MAGEMinApp
- Once installed, quite the package manager by typing 'BACKSPACE' key

Launch MAGEMinApp

- In the Julia terminal, type:
 'using MAGEMinApp'
 Then
 'App()'
- The following text will be displayed in the terminal:

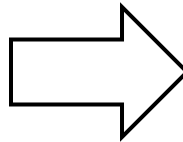
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
seph42@DESKTOP-2V82075:~$ julia -t 6

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.10.0 (2023-12-25)
Official https://julialang.org/ release

(@v1.10) pkg> add MAGEMinApp
Resolving package versions...
No Changes to `~/julia/environments/v1.10/Project.toml`
No Changes to `~/julia/environments/v1.10/Manifest.toml`

julia> using MAGEMinApp
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 1
```



- Copy and paste the address in your web-browser:

erc JGU JOHANNES GUTENBERG UNIVERSITÄT MAINZ

Reminder: Don't forget to properly cite the references used to create the diagrams by using the 'export references' buttons!

MAGEMin

Phase diagrams PTX path Isentropic path General information

Setup Diagram Trace elements

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18)

Solution phase selection: ☐

Diagram type: P-T diagram

H₂O-saturated at solidus: false

☐ Limit Ca-ox: 0.5

TE predictive model: false

Pressure [kbar]: min 0.01 max 20.01

Temperature [°C]: min 800 max 1400

Initial grid subdivision: 3 8 x 8 grid

Refinement type: Phases only

Refinement levels: 2

Buffer: no buffer

Solver: Hybrid

Verbose: none

Specific Cp: GO

Bulk-rock composition

mol%

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	fraction
SiO2	38.494
Al2O3	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K2O	0.01
Na2O	0.25
TiO2	0.1
O	0.096
Cr2O3	0.109
H2O	0

ig

General parameters

Title: KLB1 Peridotite - Anhydrous

Update Reset

Compute phase diagram

Save/Load Diagram

filename Save state Load state

State/CSV directory: /Users/nicolastiel/julia/packages/MAGEMinApp/HEFD

Help and contact

Links: Tutorials, Post Issue, Open discussion

Comments: Set of example on how to use MAGEMinApp, Something is not working properly? Need additional options?

Contributors

id=0bfcfd9-24b6-459b-b62a-9e00094b8853; MAGEMinApp GUI v0.6.2; using 12/12 threads

MAGEMin_C, the julia interface: installation



https://github.com/ComputationalThermodynamics/MAGEMin_C.jl

Install MAGEMin_C

```
julia> ]                                # opens the package
manager
pkg> add MAGEMin_C                     # MAGEMin_C
```

Load MAGEMin_C

```
julia> using MAGEMin_C                 # load MAGEMin_C
```

```
○ nicolasriel@Nicolass-MBP MAGEMinApp.jl % julia

Documentation: https://docs.julialang.org
Type "?" for help, "]"? for Pkg help.
Version 1.11.2 (2024-12-01)
Official https://julialang.org/ release

julia> ]
(@v1.11) pkg> add MAGEMin_C
Resolving package versions...
Updating `~/julia/environments/v1.11/Project.toml`
[e5d170eb] + MAGEMin_C v1.6.9
No Changes to `~/julia/environments/v1.11/Manifest.toml`

julia> using MAGEMin_C
Using libMAGEMin.dylib from MAGEMin_jll

julia> 
```